

Review of Deepfake Detection Techniques and Challenges

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Abstract—The proliferation of deepfake technology, powered by advanced generative models such as generative adversarial networks (GANs), presents challenges in digital media authenticity, public trust, and cybersecurity. We reviewed recent advancements in deepfake detection across multiple modalities, including image, video, and audio. Benchmark datasets, such as FaceForensics++, the deepfake detection challenge (DFDC), and Celeb-deepfake (Celeb-DF), have been used to develop diverse detection models. These models encompass approaches based on EfficientNet-driven transfer learning, convolutional neural network–long short-term memory (CNN-LSTM) hybrids for temporal feature extraction, graph-based neural architectures, and ensemble methods that integrate deep learning with handcrafted features. Although certain models report detection accuracies as high as 99.99% on specific datasets, many exhibit limited generalizability across different benchmarks, particularly when confronted with compression artifacts. Additionally, real-time deployment remains constrained by substantial computational demands. Emerging threats, including adversarial perturbations and diffusion-based synthetic media, necessitate the development of more resilient detection strategies. Proactive countermeasures such as blockchain-based timestamping, digital watermarking, and cryptographic hashing have been adopted to enhance media integrity. The results of the review underscore the need for lightweight, interpretable, and multimodal detection frameworks to generalize the models' applicability across diverse domains, thereby supporting reliable and scalable media verification in increasingly complex digital environments.

Keywords—deepfake detection, generative adversarial networks (GANs), convolutional neural networks (CNNs), hybrid models, spatial-temporal models

I. INTRODUCTION

Deepfakes are the synthetic media created using deep learning models, particularly generative adversarial networks (GANs), where visual or audio content is manipulated to appear convincingly authentic. These technologies generate highly realistic fake videos in which individuals appear to say or do things they never do in reality, posing a growing threat to digital authenticity and public trust. These realistic manipulations of facial expressions, voices, and body

movements have been exploited in identity theft, misinformation, and political propaganda [1,2]. As deepfakes become increasingly sophisticated and accessible, their potential for misuse expands, raising serious ethical, legal, and societal concerns.

The proliferation of deepfake technology raises ethical and legal concerns, particularly around impersonation, misinformation, and privacy infringement. The unauthorized creation of synthetic media, especially explicit or defamatory content, poses threats to individuals and institutions alike. In response, legislative efforts, such as the Deepfakes Accountability Act in the United States, criminalize malicious usage, yet enforcement remains challenging in the absence of admissible and standardized detection tools.

The psychological effect on victims is significant, ranging from anxiety to reputational damage and job loss, highlighting the need for greater institutional responsibility and increased public awareness. Platforms that host user-generated content must enforce strict policies, promote media literacy, and implement reliable detection mechanisms. Additionally, ethical AI deployment should involve transparent model behavior, fair dataset collection practices, and explainable outputs to help prevent misuse and ensure accountability.

Researchers have developed a wide array of detection methods and benchmark datasets to mitigate these threats [3]. However, performance often varies significantly due to differences in dataset characteristics, such as in healthcare [4], manipulation types, and media quality. These variations underscore the need for detection models that are robust, scalable, and explainable across diverse scenarios.

The application of deepfake detection systems is expanding across various industries [5]. In journalism, real-time detection tools help authenticate breaking news and identify disinformation campaigns [6,7]. Social media platforms, including Facebook and YouTube, are integrating artificial intelligence-based moderation systems to flag potentially manipulated content, aiding in content integrity and user safety.

In the financial sector, facial recognition systems are increasingly targeted by deepfake attacks, prompting the adoption of advanced liveness detection, multi-angle validation, and scene consistency checks. A notable case study involves telemedicine [8] and remote healthcare consultation, where verifying the authenticity of video feeds is crucial. Ensuring that a patient's identity and appearance are unaltered enhances diagnostic reliability and ensures compliance with healthcare regulations.

In this study, we conducted a structured review of current deepfake detection methods, organized by architectural approach, dataset usage, and performance. We explored the evolution from convolutional neural network (CNN)-based systems to advanced hybrid, transformer, and graph-based models, while also examining their limitations, including issues of cross-dataset generalizability and real-time feasibility. In addition, we explored emerging prevention techniques and future trends to enhance media integrity and trust.

II. METHODOLOGY

We evaluated peer-reviewed IEEE-indexed articles published between 2020 and 2025, related to deepfake detection strategies. Each study was assessed using a standardized rubric, considering model architecture, datasets used, detection approach, performance metrics, and strengths and limitations. Through the assessment, emerging trends such as transformer-based models, multimodal fusion, and adversarial robustness challenges were identified [3,6,18].

III. DEEPAKE DETECTION APPROACHES

In recent years, the rapid increase of deepfake content has sparked significant interest in developing robust detection methods [14]. Deepfake detection methods distinguish between real and synthetically generated media using a

variety of computational techniques. These methods are categorized into traditional machine learning, deep learning [15], and hybrid approaches. While early deepfake detection methods were developed based on traditional machine learning algorithms that relied on manually extracted facial or motion features, recent methods have adopted deep learning approaches capable of automatic feature extraction and representation learning.

A. CNNs

CNNs are the backbone of many deepfake detection systems due to their ability to extract spatial features from images and videos. XceptionNet, EfficientNet-B7, and custom CNN models have demonstrated high accuracy on benchmark datasets such as FaceForensics++ and Celeb-DF. EF-Net, which combines two EfficientNet-B5 models (one partially frozen and one fully trained). Models leveraging spatial artifacts and normalized facial features extracted via multi-task cascaded convolutional networks (MTCNN) have achieved up to 97.70% accuracy on the FaceForensics++ benchmark [9]. Similarly, hybrid architectures such as VGG16-CNN have demonstrated 95% accuracy on a Kaggle dataset comprising real and fake facial images, highlighting the efficacy of transfer learning and hierarchical feature extraction [7].

Despite these impressive benchmark results, CNN-based deepfake detection models have limitations in practical deployment. High computational overhead, including substantial storage and memory requirements, impedes real-time performance. Moreover, these models exhibit poor generalization when applied to unseen datasets or more sophisticated forgery techniques, resulting in notable performance degradation (Table I) [6,11].

TABLE I. SPECIFICATIONS OF CNN MODELS

Method	Model/ algorithm	Important feature	Accuracy and performance
Transfer learning-based fusion of CNNs [9]	Efficient-Fused Net (EF-Net): a fusion of two EfficientNet-B5 models (one partially frozen, one fully trained) with ReLU activation, global average pooling, dropout, and Softmax classification.	Spatial artifacts (unnatural edges, facial inconsistencies). High-level features from pre-trained EfficientNet-B5. Normalized facial frames (224×224×3) aligned via MTCNN.	FaceForensics++: 97.70% accuracy, 98.62% precision, 96.78% recall, 87.71% AUC. DFDC-Preview: 89.19% accuracy, 84.1% precision, 78.93% AUC.
Hybrid deep learning model combining transfer learning and CNN [10]	Integrated VGG16 and CNN architecture with pooling, dropout, and dense layers.	Pixel-level patterns, texture inconsistencies, and spatial features extracted via CNN and VGG16.	Achieved 95% accuracy (94% precision) with a loss of 0.1, validated through k-fold cross-validation (average 95% accuracy).
CNN-based deep learning with data augmentation [1]	CNN-based deep learning with data augmentation.	Spatial features (edges, textures) via convolutional layers. Data augmentation (rotation, flipping, GAN-generated synthetic samples).	Test Accuracy: 92.05%. AUC: 92.05%
Comparative analysis of machine learning and deep learning models [6]	SVM, CNN, Vision Transformer (ViT)	Spatial features (CNN), patch embeddings (ViT), handcrafted features (SVM)	CNN: 97.25% accuracy, 92.21% precision, 90% recall, 93% F1-score ViT: 98% accuracy, 93% precision, 93% recall, 94% F1-score SVM: 98% accuracy, 87.75% precision, 88.01% recall, 91.10% F1-score

B. Spatial-Temporal and Hybrid Models

To address the limitations of static image analysis, researchers have integrated temporal modeling with spatial feature extraction. Hybrid models, including InceptionResNetV2 + LSTM, analyze frame-level inconsistencies in videos, achieving 91.48% accuracy on Face Forensics++ in 40 epochs [12]. Another notable example is ResNet50 + LSTM, which combines spatial features from ResNet50 with temporal dependencies using long short-term

memory (LSTM), yielding 94.1% accuracy on datasets such as Deepfake Detection Challenge (DFDC) and Celeb-deepfake forensics (Celeb-DF) [14,15]. These models excel in capturing dynamic artifacts such as unnatural facial movement, mismatched lip-sync, or abnormal eye movement are the key indicators of manipulation, revealing subtle flaws introduced during deepfake generation. However, these models are now easily deployed due to high computational demands and dependence on large-scale training data (Table II) [13,15].

TABLE II. SPATIAL-TEMPORAL AND HYBRID MODELS

Method	Model/algorithm	Important feature	Accuracy and performance
Deep Learning [12]	InceptionResNetV2 (feature extraction) + LSTM (temporal analysis).	2048-dimensional feature vectors (spatial) from frames, temporal inconsistencies via LSTM.	Accuracy / Performance: 84.75% (20 epochs), 91.48% (40 epochs).
Hybrid (spatial-temporal deep learning) [13]	ResNet50: Extracts spatial features from frames. LSTM: Analyzes temporal dependencies across frames.	Spatial: Facial landmarks, textures (via ResNet50). Temporal: Frame-sequence anomalies (via LSTM).	Accuracy: 94.1%. AUC: 0.936. Confusion Matrix: High true positives (TP) and true negatives (TN).
Hybrid deep learning (CNN + LSTM) [12], [14]	ResNeXt50_32x4d (CNN backbone) + LSTM (256 hidden units)	Spatial features (ResNeXt50) Temporal dependencies (LSTM) Facial ROIs (cropped and segmented)	FF++: 96.67% Accuracy, 96.68% AUC Celeb-DF V1: 91.80% Accuracy, 92.00% AUC
Hybrid deep learning (spatial-temporal analysis) [15]	ResNet50 (for spatial feature extraction) + LSTM (for temporal sequence processing)	Frame-level spatial features (e.g., facial artifacts, inconsistencies) and temporal patterns (e.g., frame sequences)	Accuracy: 94.1%

C. Graph-Based and Transformer Approaches

Recent advancements include graph-based and transformer architectures, which offer superior generalization and interpretability. GatedGCN, paired with EfficientNet-b0, models region-wise facial inconsistencies (e.g., eyes, nose, mouth) as graph nodes, achieving an area under the curve (AUC) of 98.7% on FMFCC-V and robust cross-dataset performance (a 89.4% AUC on Celeb-DF) [16]. Transformer-

based methods such as Cross-ViT + DenseNet leverage multi-scale attention to fuse global and local features, attaining state-of-the-art results (an AUC of 99.99% on DeeperForensics 1.0) [17]. Other models, such as spatial kernel selection and halo attention network (SKSHA-Net), dynamically adjust spatial receptive fields to detect subtle artifacts, achieving a 99.32% AUC on FaceForensics++ HQ [18]. Despite their success, these models are computationally intensive and may struggle with low-quality or compressed media [19,20].

TABLE III. GRAPH-BASED AND TRANSFORMER MODELS

Method	Model/algorithm	Important feature	Accuracy and performance
Graph-based deep learning [16]	Graph Neural Network (GatedGCN) with EfficientNet-b0 for feature extraction	Spatial-temporal correlations between facial regions (eyes, nose, mouth, rest), extracted via OpenFace2.0 and cross-attention mechanisms	Highest AUC (98.7%) on FMFCC-V, demonstrating robustness to Asian face forgeries. Strong cross-dataset generalization (e.g., 89.4% AUC on Celeb-DF without fine-tuning).
Hybrid deep learning (CNN + Transformer) [17]	DenseNet121 + Cross-ViT (Cross Vision Transformer)	Spatial features (DenseNet), global context (Cross-ViT), multi-scale attention	AUC: 99.99% (DeeperForensics 1.0), 97.4% (Celeb-DF); F1-score: 99.0% (DeeperForensics 1.0), 95.1% (Celeb-DF)
Deep learning-based detection using frequency domain features [18]	Frequency Enhanced Self-Blended Images (FSBI) with EfficientNet-B5	Discrete Wavelet Transform (DWT) coefficients from self-blended images (SBI)	AUC of 95.49% on Celeb-DF (cross-dataset) and 95.13% on FF++ (within-dataset)
Deep learning-based detection with dynamic spatial and attention mechanisms [19]	Spatial Kernel Selection and Halo Attention Network (SKSHA-Net)	Multi-scale spatial receptive fields (SKS) and localized pixel relationships (HA)	Intra-testing: 97.16% Acc (FF++ HQ), 99.32% AUC (FF++ HQ), 94.75% EER (Celeb-DF) Cross-testing: 70.54% AUC (Celeb-DF), 69.58% AUC (WildDeepfake)
Deep learning-based detection with facial landmark guidance [20]	Trail Tracing Network (TTNet)	Facial landmarks and RGB data, enhanced by Trace Enhancement Module (TEM)	In-dataset (FF++ HQ): Avg AUC of 87.6% (best: 99.9% for NeuralTextures) Cross-dataset: Avg AUC of 83.6% (94.0% on Celeb-DF, 87.5% on DFDCp)

IV. DATASET AND PERFORMANCE

A thorough understanding of the datasets used in deepfake detection research is crucial for evaluating the effectiveness and generalizability of proposed models. The performance of detection algorithms varies significantly depending on the dataset used for training and testing, as each dataset contains different types of manipulations, quality levels, and facial variations [20]. Moreover, while several models perform well within a single dataset, they might struggle to maintain accuracy across different datasets or under challenging conditions. We analyzed commonly used deepfake datasets and evaluated intra-dataset performance to explore cross-dataset generalization issues and the robustness of detection methods in adversarial scenarios.

A. Common Datasets and Intra-Dataset Performance

The most widely used datasets include FaceForensics++, DFDC, Celeb-DF, and DeeperForensics 1.0, which provide diverse samples of manipulated images and videos. Models trained and tested on the same dataset often achieve high performance. For instance, SFSimNet reached a 99.18% AUC on FaceForensics++ (HQ) [21] while AMTENnet achieved a 98.52% accuracy on the same dataset [22]. However, intra-dataset results can be misleading, as models overfit to specific artifacts or manipulation techniques [3,23].

B. Common Cross-Dataset Generalization Challenges

A significant limitation in deepfake detection is a decrease in performance when models are tested on unseen datasets. For example, frequency-enhanced self-blended images (FSBI), which combine SBI with discrete wavelet transform (DWT), achieved a 95.49% AUC on Celeb-DF but struggled with NeuralTextures due to unique artifact patterns [18]. Similarly, the temporal transformer network (TTNet), which leverages facial landmarks for detection, showed an average cross-dataset AUC of 83.6%, highlighting the difficulty of generalizing across diverse manipulation methods [20]. High-frequency diversified augmentation (HFDA) mitigates this issue by perturbing high-frequency components in the Fourier domain, improving cross-dataset AUC to 85.09% on Celeb-DF-v2 [24].

C. Robustness to Adversarial Conditions

Deepfake detectors must also contend with adversarial attacks, compression artifacts, and low-resolution inputs. A multi-scale graph attention transformer (MSG-GAT), a spectral-guided graph attention network, demonstrated resilience to adversarial conditions, achieving a 99.3% accuracy on Celeb-DF [25]. Conversely, vision transformer (ViT) + EfficientNet models exhibited performance degradation on low-quality videos, underscoring the need for robust preprocessing and adaptive architectures [26]. The lack of standardized evaluation protocols further complicates comparisons, as AUC and accuracy vary based on dataset splits and manipulation types [23,27].

V. STANDARDIZATION AND BENCHMARKING

While numerous datasets and evaluation metrics exist, the lack of universal benchmarking standards hampers model comparability and reproducibility. Discrepancies in video resolution, compression artifacts, manipulation techniques, and data partitioning lead to inconsistent and misleading performance metrics. For instance, a model trained on high-

quality synthetic videos fail on real-world compressed videos sourced from social media.

To address this, the unified evaluation protocols and a centralized leaderboard are required to assess detection models across diverse datasets. Incorporating advanced metrics such as compression robustness, adversarial resistance, demographic fairness, and real-time latency enables an understanding of system performance. These benchmarks improve research rigor to build confidence among stakeholders adopting detection technologies.

VI. PREVENTION OF DEEPFAKE

Several studies have proposed prevention mechanisms to authenticate media and counteract the malicious use of deepfakes. Blockchain timestamping [28], watermark injection [28], and cryptographic hashing [29] are considered in verifying the originality and integrity of media content. However, these approaches face limitations in terms of scalability, susceptibility to anti-forensic attacks, and the need for cross-platform adoption. As a result, their effectiveness remains constrained in real-world settings.

Emerging trends indicate a shift toward proactive and intelligent detection systems that integrate explainable artificial intelligence (XAI), federated learning, and real-time deployment optimization. Models leveraging vision transformers, GAN discriminators, and multimodal biometric analysis are gaining prominence due to their enhanced capability to detect and localize manipulated regions with greater precision [10,17,22]. These advancements enable the development of more resilient and adaptive detection frameworks, capable of effectively responding to the rapid evolution of deepfake generation techniques.

VII. CONCLUSION

Despite remarkable advancements in deepfake detection, challenges persist. Current methods, particularly deep learning models, often struggle with generalization, high computational cost, and real-time deployment feasibility. While hybrid and ensemble models demonstrate superior accuracy, they are often hindered by increased complexity and latency. Moreover, low-quality or adversarial manipulations continue to expose vulnerabilities in existing systems. Further research is necessary to develop adaptive, lightweight, and interpretable detection models supported by standardized benchmarks and robust cross-domain evaluation protocols. Interdisciplinary collaboration and timely regulatory measures are essential in ensuring secure and trustworthy AI-driven media authentication.

To ensure long-term success in deepfake detection, the current limitations in generalizability, efficiency, explainability, and reproducibility of existing models must be addressed.

First, cross-dataset training and evaluation must be prioritized, as generalization across datasets remains one of the most pressing challenges in deepfake detection. Models trained on specific datasets often fail to adapt to unfamiliar manipulation techniques and real-world conditions. Although many detection models achieve high accuracy on their training datasets, performance degrades when applied to unseen data with differing characteristics. To address this limitation, cross-dataset evaluation must become a standard practice to ensure robustness in unpredictable environments. A lack of generalizability can be mitigated through domain adaptation

techniques, including adversarial training, style transfer, and data augmentation, which enable models to learn invariant features across heterogeneous data distributions. Additionally, multimodal fusion, integrating visual, audio, and textual cues, has demonstrated promise in improving detection accuracy across domains [30,31].

Second, researchers must develop low-latency architectures for mobile and edge deployment. Practical deepfake detection requires models to operate efficiently on resource-constrained devices such as smartphones, surveillance systems, and embedded platforms. Model quantization, pruning, and neural architecture search (NAS) can be used to significantly reduce computational complexity while preserving performance. Lightweight models that maintain high accuracy and enable fast inference are essential for real-time detection and scalability. DeepGuardNet, specifically designed for edge deployment, exemplifies the feasibility of efficient real-time detection [32,33].

Third, the integration of XAI is critical. Deep learning models are criticized for their lack of interpretability, functioning as "black boxes." Explainability modules, such as attention heatmaps, feature attribution maps, and saliency visualizations, enhance transparency and facilitate an understanding of model behavior. This is particularly important in legal and forensic contexts, where model decisions must be justified in formal proceedings. Techniques like Grad-CAM and saliency visualization offer valuable insights into decision-making processes [34,35].

Fourth, future systems must adopt human-AI collaboration. While AI excels at processing large volumes of data, human judgment remains indispensable in scenarios requiring nuanced interpretation. Hybrid workflows that combine automated detection with manual verification yield more reliable outcomes, especially in sensitive domains such as journalism, legal evidence validation, and psychological assessment. Interfaces that support interactive feedback and user-in-the-loop mechanisms will enhance trust, usability, and adoption.

Finally, open-source frameworks are required to foster reproducibility and transparency. Publicly available codebases, benchmark datasets, and evaluation results enable researchers to validate results, compare methodologies, and build upon previous results. The adoption of standardized evaluation protocols and shared repositories accelerates innovation and reduces redundant efforts. The findable, accessible, interoperable, and reusable (FAIR) principles in data sharing must be employed to cultivate a collaborative and sustainable ecosystem for deepfake research.

Collectively, these recommendations help to bridge the gap between academic innovation and practical deployment, ensuring that deepfake detection systems are robust, efficient, interpretable, and accessible across a wide range of applications.

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